Statistical Methods II: Week 9 Assignment (due 24 March, 2025)

1. Derive the Method of Moments estimator of the rate parameter of the Poisson distribution.

2. Derive the Method of Moments estimators of the mean and the variance parameters of the Poisson distribution.

3. Derive the Method of Moments estimators of the mean and the variance parameters of the Gamma distribution.

4. Derive the Method of Moments estimators of the shape and the scale parameters of the Gamma distribution.

5. Suppose the random variables $Y, X_1, X_2, \ldots, X_k$ have mean vector μ and varaince-covariance matrix Σ, partitioned as

$$\mu = \begin{pmatrix} \mu_Y \\ \mu_X \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \sigma_Y^2 & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix},$$

   where μ_Y and σ_Y^2 are the mean and the variance of Y. What are the method of moments estimators of $\beta = \Sigma_{XX}^{-1}\Sigma_{XY}$ and $\beta_0 = \mu_Y + \beta^T(X - \mu_X)$, where X is the column vector with elements $X_1, X_2, \ldots, X_k$? You may assume that the joint distribution of the random variables are defined with any restriction on the matrix Σ, except that it is positive definite.

6. Suppose $\varepsilon_{Y|X} = Y - \mu_Y - \Sigma_{YX}\Sigma_{XX}^{-1}(X - \mu_X)$, where the notations and assumptions of Question 5 are used. Suppose further that for another random variable Z, we similarly define $\varepsilon_{Z|X}$. What is the best linear prediction of $\varepsilon_{Y|X}$ in terms of $\varepsilon_{Z|X}$, which minimizes the mean squared error of prediction?

7. In Question 6, what is the partial correlation of Y and Z, after adjustment for the linear effects of X?